

Match-Regist assumes that every speaker appearing in the ground truth has been registered in advance, same as SA-ASR. The model is required to correctly assign each speaker to the corresponding name. This setting closely reflects scenarios where known users are pre-registered, and personalized outputs (e.g., transcripts with speaker names) are desirable. For Match-Regist, accurate speaker-name association becomes essential, and the model must effectively leverage the registered speaker information for identification.

Over-Regist refers to the case where more speakers are registered than actually appear in the audio. In this scenario, the model must determine which registered speakers are not present in the current utterance, and correctly perform speaker-attributed SDR for the remaining active speakers. This setup is more challenging than Match-Regist, as the model must handle redundant speaker information and suppress irrelevant identities. It also aligns well with practical scenarios, where a large pool of users may be pre-registered, but only a subset participate in a given interaction.

Overall, the proposed speaker registration framework allows SpeakerLM to flexibly perform SDR under different speaker supervision levels, making it applicable to various practical multi-speaker tasks, from anonymous transcription to personalized speaker transcription.

3.3 Multi-Stage Training Strategy

We apply a four-stage training strategy to progressively enhance the model’s audio understanding capabilities.

Stage 1. To enhance the ASR performance, we first train SpeakerLM using only ASR data, leading to an ASR model, denoted by **SpeakerLM-ASR**. To efficiently adapt the LLM while preserving its pre-trained language capabilities, we apply a Low-Rank Adaptation (LoRA) (Hu et al. 2022) strategy. Following the ASR training setup of MinMo (Chen et al. 2025a), we sample 600,000 hours of audio from publicly available ASR datasets for training. In this stage, the speaker embedding extractor and projector are not included in the model. In the subsequent three stages, these modules are incorporated into the whole architecture.

Stage 2. We train the randomly initialized projectors using simulated SDR data while keeping both the LLM and audio encoder frozen, which aims to rapidly align the audio and text for SDR domain. Training on simulated data allows the projectors to establish an initial coarse alignment under a simplified distribution. Compared with real recordings, the simulated mixtures are created by simply concatenating utterances from different speakers, without modeling very strong noise or reverberation.

Stage 3. In the remaining two stages, we use real SDR data for training, to further adapt the model to practical acoustic conditions and speaker diversity. In this stage, we jointly fine-tune the audio encoder and the projectors, while keeping the LLM frozen, which allows the model to better capture acoustic patterns and characteristics.

Stage 4. Finally, we jointly fine-tune all modules while applying LoRA to the LLM. This stage enables jointly optimization to better integrate linguistic and acoustic information in complex multi-speaker SDR tasks.

4 Data Composition

4.1 Real Data Collection

In this work, we conduct experiments on Mandarin SDR datasets. Overall, we sample 238.55 hours of audio clips from public corpora for training and evaluation, covering various real-world multi-speaker scenarios. In addition, we use 7456.99 hours of in-house data for training and validation, to further enhance the model performance. Detailed statistics are summarized in Table 1.

Specifically, the public Alimeeting (Liang et al. 2023) and AISHELL4 (Fu et al. 2021) datasets are collections of multi-speaker conversation recordings in meeting rooms, including both far-field overlapping audio clips and near-field recordings for each speaker. Following previous works (Han et al. 2025a; Chen et al. 2025b), we use the first channel of the far-field signals for training and evaluation. The in-house data contains multi-speaker audio samples recorded in professional studios. The AISHELL5 dataset (Dai et al. 2025) was recorded using far-field microphones in a hybrid electric car. The dataset includes diverse external noise sources such as wind and tire, along with internal noises from music and air conditioning, making it a more challenging setting than standard meeting room or studio environments. The dataset has two official test set splits for ASR and SDR tasks respectively. We use the first channel of the far-field audio from the SDR test split as the test set in this work, denoted by AISHELL5-Eval. Notably, AISHELL5-Eval presents a challenging acoustic condition that is significantly different from the domains of the training data, hence it is primarily used to evaluate the out-of-domain SDR performance (**generalizability**) of the proposed SpeakerLM.

Dataset	Duration (h)	# Spk	Usage
AliMeeting-Train	104.75	2~4	Train
AISHELL4-Train	107.50	4~8	Train
In-House-Train	7,426.70	2~7	Train
In-House-Valid	30.29	2~7	Valid
AliMeeting-Eval	10.00	2~4	Test
AISHELL4-Eval	12.72	5~7	Test
AISHELL5-Eval	3.58	2	Test

Table 1: Statistics of the real SDR data, where “#Spk” denotes the number of speakers in each recording.

4.2 Simulated Data Creation

As described in Section 3.3, we apply a four-stage training strategy to train SpeakerLM. In the second stage, we use simulated data for quickly training the projectors and audio-text alignment. Specifically, we use the near-field speech samples from AliMeeting-Train, AISHELL2 (Du et al. 2018), Librispeech (Panayotov et al. 2015) and In-House-Train for SDR data simulation. The near-field speech samples are randomly mixed to simulate multi-speaker conversation segments, with each segment lasting 50 seconds and containing 2 to 4 speakers. Following previous work (Landini et al. 2022), we randomly add real-world noises and reverberations to the clean speech samples, with signal-to-noise ratios (SNRs) uniformly sampled from [10dB, 20dB].

In total, the simulated data comprise 5,000 hours audio samples for training and 5.6 hours for evaluation; the test split is denoted by Simulation-Test.

5 Experiments

5.1 Implementation Details

The majority of audio samples are recorded at 16 kHz and any samples with a different sampling rate are resampled to 16 kHz to ensure uniformity. Recordings are randomly split into clips ranging from 40 to 50 seconds in duration to train and test SpeakerLM. For speaker registration, the speech of the registered speaker is segmented into 2-10 seconds clips for embedding extraction, and the corresponding embeddings are averaged to generate a single representative speaker embedding. For Over-Regist, the number of over-registered speakers (i.e., N_{ov} in Eq. 1) uniformly ranges from 1 to 50 during training.

We use the AdamW (Zhou et al. 2024) optimizer for optimization. The learning rate is initially set to $1e-5$, linearly increased to $5e-5$ during the warm-up phase, and then gradually decayed following a cosine schedule. We use a dynamic batching strategy, adjusting the number of samples per batch based on their length, with a maximum token limit of 6K to balance efficiency and memory usage. Models are trained on 4 NVIDIA A800 GPUs for 1M steps at each stage, with a validation step of 10K. The training pseudocode and the specific prompts used for LLMs are presented in code page.

5.2 Evaluation Setups

Evaluation Metrics. We evaluate the SDR performance on the commonly used publicly available benchmarks, namely, the in-domain AliMeeting-Eval and AISHELL4-Eval, and the out-of-domain AISHELL5-Eval. Following previous works (Liang et al. 2023; Wang et al. 2024; Dai et al. 2025), we use Character Error Rate (CER), concatenated minimum permutation CER (cpCER), Δ_{cp} , speaker-attributed CER (saCER), and Δ_{sa} as evaluation metrics.

CER measures ASR performance by comparing the predicted transcripts against the ground-truth text, ignoring speaker identities. In contrast, cpCER and saCER jointly evaluate both ASR and SD, calculated by comparing the predicted speaker-attributed transcripts against the ground-truth transcripts assigned to each speaker, hence reflecting the overall system performance for the SDR task. Specifically, for the no-speaker-registration scenario, cpCER is computed after conducting the speaker alignment between the reference and predicted speaker-attributed transcripts through minimum edit distance, which finds the optimal permutation of speaker labels to minimize the CER. For the registered-speaker scenario where speaker identities are known in advance, saCER is computed by directly aligning the predicted and reference transcripts based on speaker names, without performing permutation search. Δ_{cp} denotes the difference between cpCER and CER:

$$\Delta_{cp} = \text{cpCER} - \text{CER} \quad (2)$$

Δ_{cp} reflects the performance drop due to speaker attribution errors, hence serving as a measure of the SD performance.

Similarly, Δ_{sa} is calculated as:

$$\Delta_{sa} = \text{saCER} - \text{CER} \quad (3)$$

All metrics are measured in percentage. Lower values represent better performance. We use **cpCER and saCER as the primary metrics** since they provide a comprehensive measure of both SD and ASR performance.

Baseline Systems. Figure 1 depicts three different frameworks for SDR, i.e., SD+ASR, SD+ASR+LLM, and E2E-SDR. The proposed SpeakerLM belongs to the E2E-SDR paradigm, where a unified model is applied for all tasks in an end-to-end manner. We build four baseline systems for the SD+ASR paradigm using SOTA open-source models, and apply competitive pre-trained LLMs to correct the outputs from SD+ASR systems, leading to three different SD+ASR+LLM baselines.

For the SD+ASR paradigm, we use Paraformer-large (Gao et al. 2022) as the ASR module, which was pre-trained on 60K hours of audio data and achieves SOTA performance on various Mandarin ASR benchmarks. For the SD module, we apply four SOTA open-source SD toolkits, including 3D-Speaker (Chen et al. 2025b), Pyannote 3.1 (Bredin 2023), Diarizen-base (Han et al. 2025a), and Diarizen-large (Han et al. 2025b). Among the four SD models, 3D-Speaker and Diarizen-large have achieved SOTA performance on different benchmarks, while Pyannote 3.1 and Diarizen-base demonstrate competitive performance. By combining the four SD systems with Paraformer-large, we obtain four SD+ASR baseline systems, namely, **3D-Speaker+Para**, **Pyannote+Para**, **Diarizen-base+Para**, and **Diarizen-large+Para**.

SD+ASR+LLM refers to a pipeline where a text LLM is used to correct the outputs from the SD+ASR system. For SD+ASR+LLM baselines, prior works using a text LLM for SD+ASR correction are mainly based on English datasets (Wang et al. 2024; Park et al. 2024; Efstathiadis, Yadav, and Abbas 2025). Following the paradigm, we adapt Qwen2.5-7B-Instruct (Team 2024) and ChatGPT4.5 (Singh 2025) for correction. We first evaluate their zero-shot performance (z.s.) using prompts following prior work (Efstathiadis, Yadav, and Abbas 2025), then fine-tune Qwen2.5-7B-Instruct (f.t.) on our training datasets for SD+ASR correction.

5.3 Performance without Speaker Registration

We first evaluate the performance of SpeakerLM without speaker registration, where each speaker in the prediction results is represented by an anonymous ID.

Comparison with Baselines. Table 2 compares our SpeakerLM and the baselines under **No-Regist** condition on various test sets. For SD+ASR with four models included, Diarizen-large+Para achieves the best performance. Compared to 3D-Speaker, both Pyannote and Diarizen employ a neural speaker activity prediction model to replace the VAD module, enabling more effective handling of overlapping speech. Building upon the Pyannote pipeline, Diarizen series further enhances the SD performance by incorporating acoustic features from the pre-trained WavLM model.

For SD+ASR+LLM baselines pipelined with five models, we incorporate a pre-trained text LLM to correct the mis-

System	Params	#Models	Alimeeting-Eval			AISHELL4-Eval			AISHELL5-Eval		
			CER↓	cpCER↓	Δcp ↓	CER↓	cpCER↓	Δcp ↓	CER↓	cpCER↓	Δcp ↓
SD+ASR											
3D-Speaker+Para	70M	4	21.30	24.94	3.64	23.02	26.01	2.99	60.16	64.12	3.96
Pyannote+Para	70M	4	21.30	24.45	3.15	23.02	28.22	5.20	60.16	68.37	8.21
DiariZen-base+Para	95M	4	21.30	23.97	2.67	23.02	27.27	4.25	60.16	66.89	6.73
DiariZen-large+Para	140M	4	21.30	23.20	1.90	23.02	25.78	2.76	60.16	61.81	1.65
SD+ASR+LLM											
ChatGPT4.5 (z.s.)	-	5	30.63	38.64	8.01	33.02	39.21	6.19	67.34	79.05	11.71
Qwen2.5-7B-Instruct (z.s.)	7B	5	40.10	51.01	10.91	33.92	44.16	10.24	65.67	73.30	7.63
Qwen2.5-7B-Instruct (f.t.)	7B	5	21.38	22.65	1.27	23.05	24.93	1.88	60.17	61.63	1.46
E2E-SDR											
SpeakerLM-ASR	7B	1	20.98	-	-	18.02	-	-	55.50	-	-
SpeakerLM (212.25h)	7B	1	18.63	32.22	13.59	17.75	26.14	8.39	48.40	64.96	16.56
SpeakerLM (694.06h)	7B	1	18.14	29.60	11.46	17.48	25.28	7.80	48.39	54.81	6.42
SpeakerLM (2,269.57h)	7B	1	17.32	27.97	10.65	17.32	23.10	5.78	47.78	50.04	2.26
SpeakerLM (7,638.95h)	7B	1	13.97	16.05	2.08	17.17	18.37	1.20	47.24	47.81	0.57

Table 2: The performance of our SpeakerLM (with different amounts of training data) and the baselines on the *in-domain* AliMeeting-Eval, AISHELL4-Eval, and the *out-of-domain* AISHELL5-Eval set. The best results are boldfaced. For SD+ASR+LLM, we use DiariZen-large+Para as the SD+ASR front-end, where “z.s.” is “zero shot” and “f.t.” is “fine-tune”. All SpeakerLM variants are evaluated under the *No-Regist* condition. “#Models” represents the number of models included.

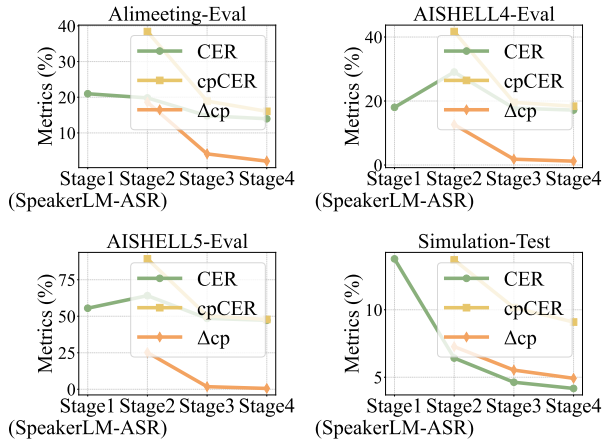


Figure 4: The performance of SpeakerLM on test sets under *No-Regist* condition across different training stages.

assigned speaker labels in the results from the best SD+ASR system (i.e., Diarizen-large+Para). Results show that both Qwen2.5-7B-Instruct and ChatGPT-4.5 perform poorly in zero-shot SDR correction, even degrading the overall performance. This finding is consistent with observations in prior works (Park et al. 2024; Efstathiadis, Yadav, and Abbas 2025). This is mainly due to hallucination of LLMs: despite that we explicitly instruct the LLM to only modify speaker labels, the LLM often alters the content of the speakers’ utterances, leading to a significant increase in CER. Finetuning Qwen2.5-7B-Instruct substantially mitigates the hallucination problem, leading to slight SDR performance improvement over Diarizen-large+Para.

Different from the cascaded architectures of SD+ASR and SD+ASR+LLM, the proposed SpeakerLM is a unified E2E-SDR model that jointly performs speaker diarization and recognition. To explore how much SpeakerLM benefits from the ASR training stage, we first report the CER of SpeakerLM-ASR, which is trained on the ASR data mentioned in Section 3.3. Compared with Paraformer-large, SpeakerLM-ASR is trained on a larger ASR dataset and thus achieves lower CER across all test sets, which naturally contributes to a lower cpCER for SpeakerLM. However, it should be noted that Δ cp is not affected by the model’s ASR performance, which serves as a reliable metric for evaluating the speaker attribution performance.

Initializing SpeakerLM by SpeakerLM-ASR, we conduct the following training with different scales of SDR data, ranging from 212.25 hours (Alimeeting-Train+AISHELL4-Train) to 7,638.95 hours. Results show that when trained on a limited amount of SDR data, SpeakerLM falls behind most cascaded baselines. However, as the training data scale up, SpeakerLM demonstrates a strong data scaling capability, with substantial improvements in cpCER and Δ cp. Moreover, we observe that the improvements on CER are marginal. This is because the in-house data mainly consists of near-field recordings, which offer limited ASR gains for far-field speech with reverberation. With 7,638.95 hours training data, SpeakerLM significantly outperforms all baselines. In terms of cpCER, SpeakerLM achieves absolute improvements of 6.60%, 6.56%, and 13.82% over the strongest cascaded system on AliMeeting-Eval, AISHELL-4-Eval, and AISHELL-5-Eval, respectively. Notably, SpeakerLM also achieves superior performance on the out-of-domain and challenging test set, AISHELL5-Eval, with a Δ cp of **0.57**, indicating **strong robustness and generalizability of SpeakerLM to unseen and noisy acoustic environments**.